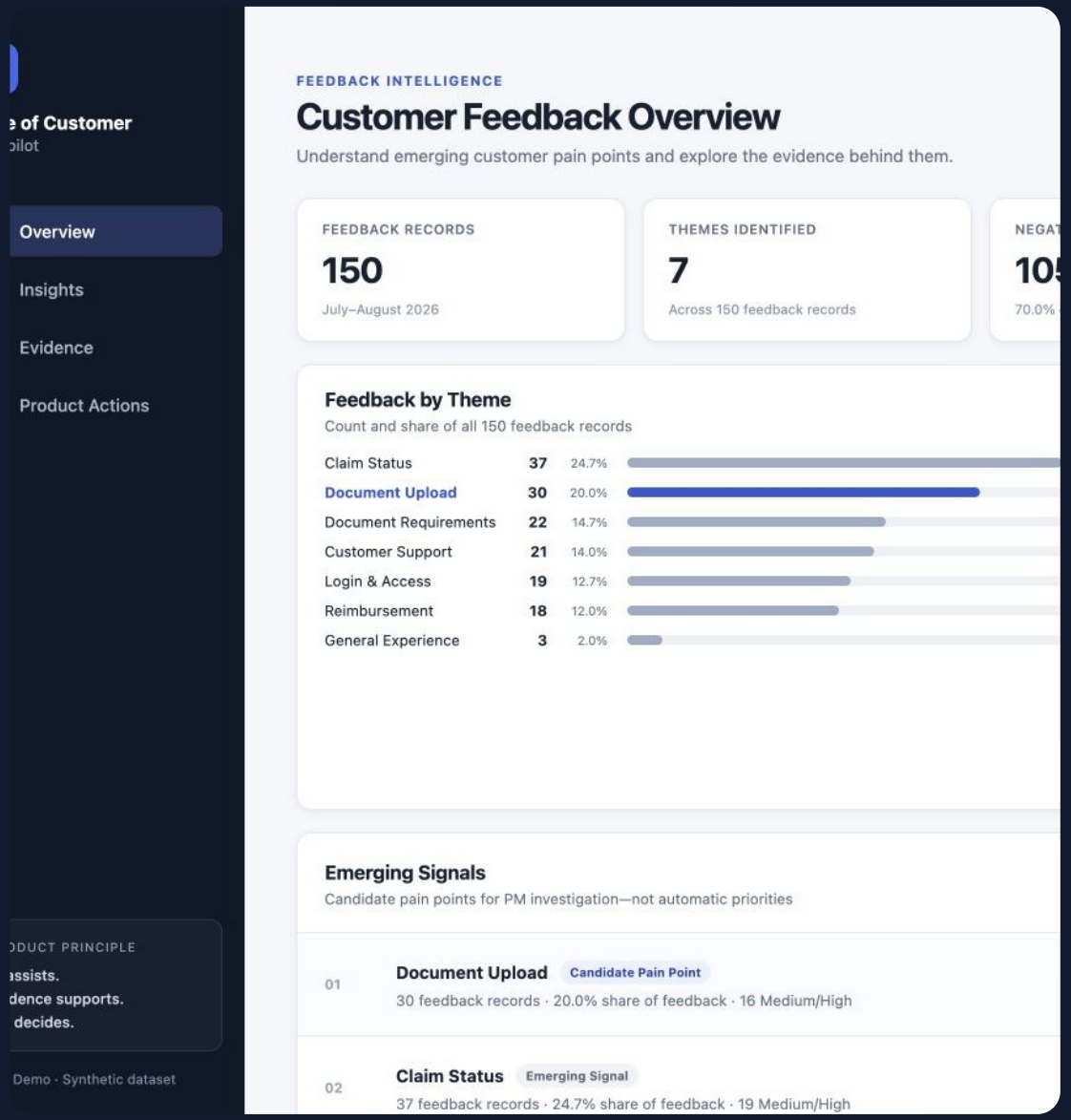
AI PRODUCT MANAGEMENT CASE STUDY

Voice of Customer AI Copilot

Turning unstructured insurance feedback into traceable product signals

SYNTHETIC DATASET

Product strategy · Prototype · AI evaluation



CASE STUDY

Product Managers receive feedback as text, not decisions

Feedback arrives across channels with different wording and levels of detail.

01 **Which product area is involved?**

02 **What kind of feedback is this?**

03 **How much impact does the text describe?**

04 **Which records support the pattern?**

RAW FEEDBACK

“Did the file arrive?”

“Why did my status move back?”

“Which receipt do I need?”

“Please add separate profiles.”

The opportunity is faster evidence review, not automated prioritization.

Product opportunity and value hypothesis

PRODUCT HYPOTHESIS

If PMs can identify recurring friction and trace it back to customer evidence faster, they can spend less time organizing feedback and more time investigating the problems that deserve attention.

Faster synthesis

Less manual effort organizing feedback before analysis.

Earlier visibility

Recurring friction becomes easier to spot across many records.

Evidence-backed investigation

Patterns remain linked to customer feedback before a PM acts.

VALUE HYPOTHESIS

To be validated with real PM workflows and production data

Primary user and job to be done

PRIMARY USER

Product Manager

Owns the product decision and the investigation that follows.

JOB TO BE DONE

When reviewing a large feedback set, help me find recurring friction and inspect the evidence so I can decide what needs deeper investigation.

AI organizes the evidence

PM owns the decision

Customer Service Manager remains a possible future user, not part of the focused MVP story.

Product workflow

01
Customer feedback

02
AI classification

03
Structured signals

04
Candidate pain point

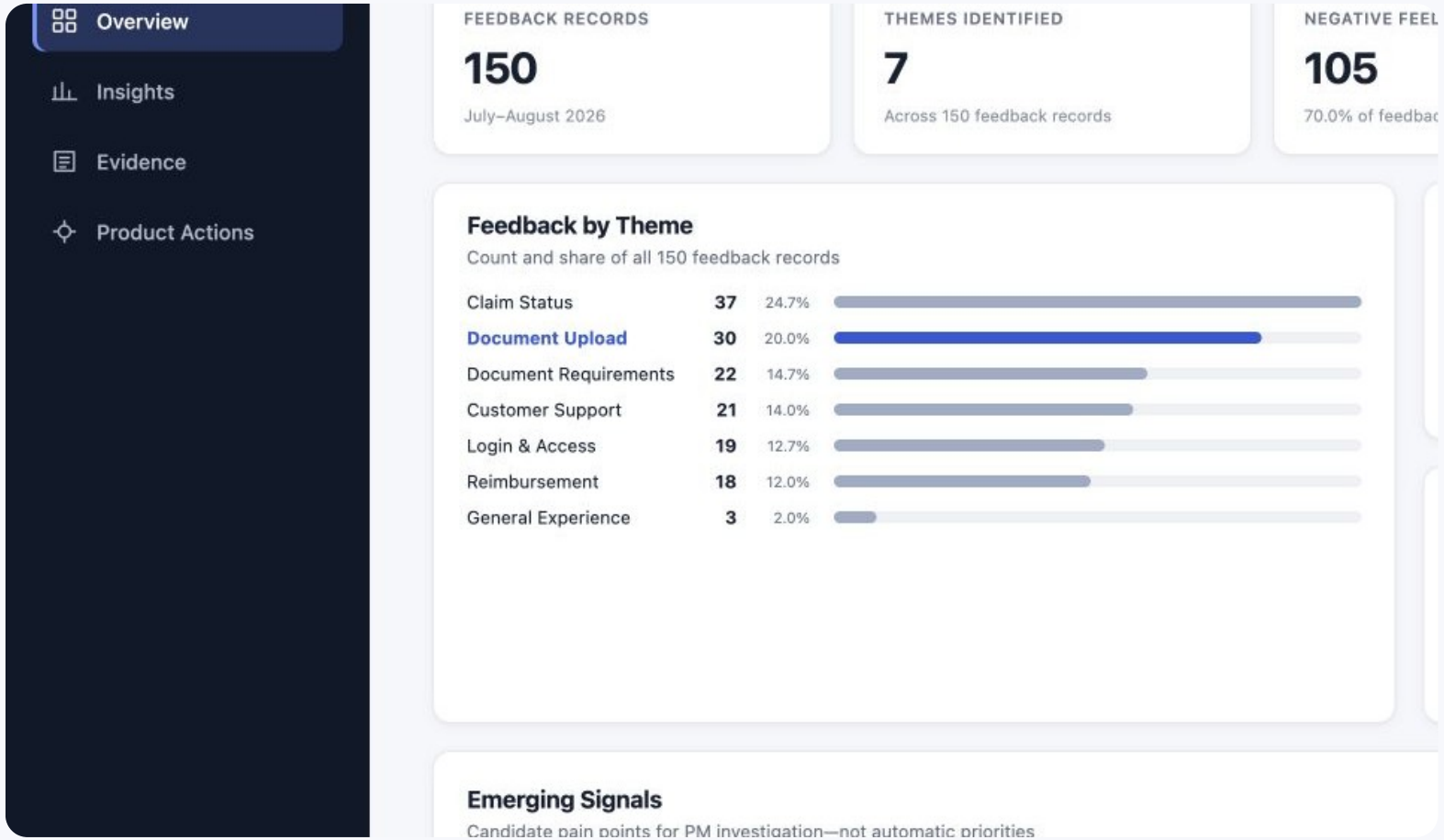
05
Customer evidence

06
PM investigation

AI structures and surfaces signals

The Product Manager decides what deserves action

The prototype makes every insight traceable



01 **Overview**

Ranks candidate signals

02 **Insight**

Separates volume and share

03 **Evidence**

Returns to original records

04 **Product action**

Keeps the PM responsible

WORKFLOW PROTOTYPE

Earlier seven-theme UI taxonomy

CASE STUDY

Four dimensions create more useful product signals

Theme

What product area is involved?

Document Upload

Feedback Type

What is the customer trying to communicate?

Technical Issue

Sentiment

What attitude is expressed?

Negative

Severity

What impact is described?

Medium

Separating subject, intent, attitude, and impact prevents one label from carrying four meanings.

CASE STUDY

Development and holdout served different purposes

DEVELOPMENT LANE

150	synthetic records
40	human-reviewed development records
V1 · V2 · V3	prompt experiments
V1	final candidate lock

FINAL HOLDOUT

24

locked records

Selected candidate evaluated once on unseen examples

NO POST-HOLDOUT TUNING

CASE STUDY

More prompt detail did not improve overall performance

Development set · n = 40

Prompt	Theme	Type	Sentiment	Severity	Exact
V1	**97.5%**	**85.0%**	**82.5%**	**92.5%**	**67.5%**
V2	95.0%	90.0%	52.5%	82.5%	37.5%
V3	95.0%	87.5%	85.0%	85.0%	62.5%

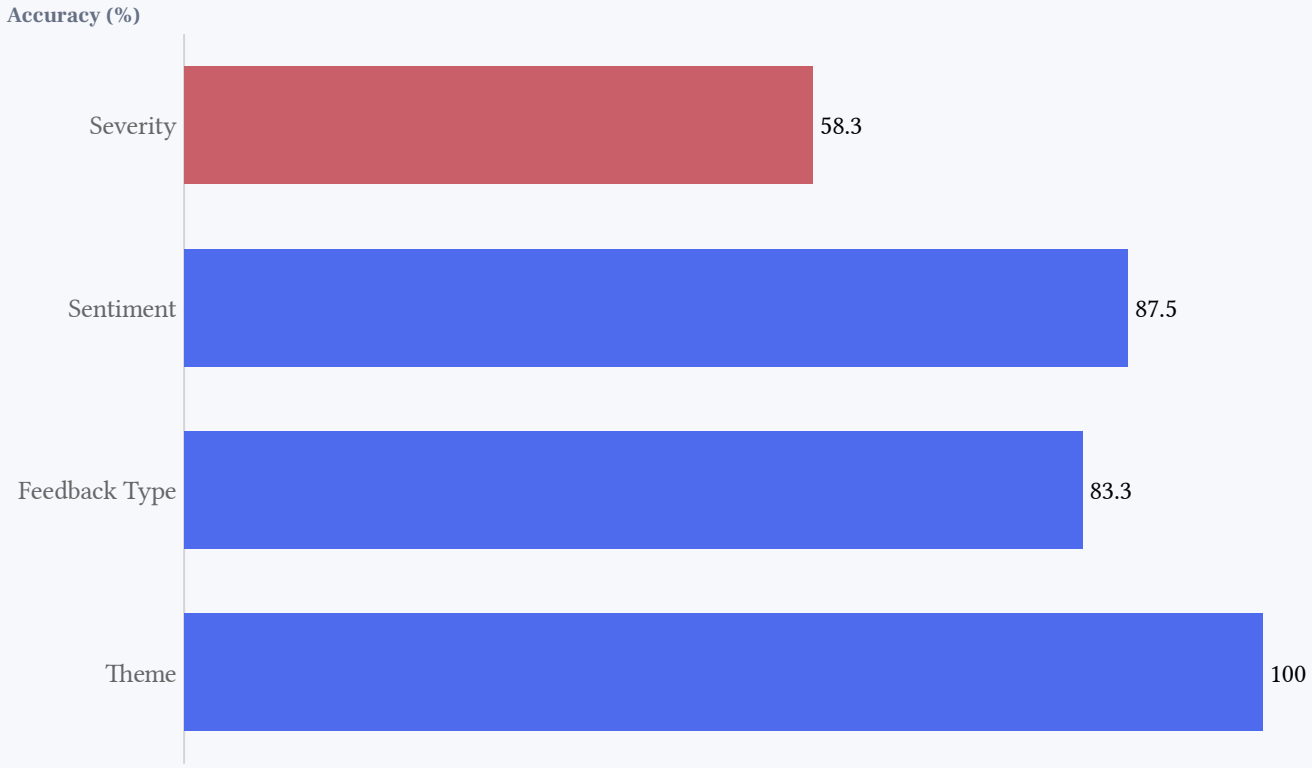
V2 improved Feedback Type but caused major regressions. V3 recovered much of the loss, yet V1 retained the strongest overall balance.

FINAL CANDIDATE: BASELINE V1

CASE STUDY

The unseen holdout revealed uneven reliability

Final holdout · n = 24



Theme 100.0% Feedback Type 83.3% Sentiment 87.5% Severity 58.3%

11 / 24

45.8% all-fields exact match

A record counts as exact only when all four fields agree.

45.8% is not a single overall ‘AI accuracy.’

CASE STUDY

Severity was the weakest signal on unseen data

Human label \ Gemini prediction	No Issue	Low	Medium	High
No Issue	3	7	0	0
Low	1	3	2	0
Medium	0	0	6	0
High	0	0	0	2

7 of 10

No Issue records predicted Low

The largest failure pattern

Medium: 6 of 6 correct

High: 2 of 2 correct

The main boundary problem was no stated impact versus limited impact.

CASE STUDY

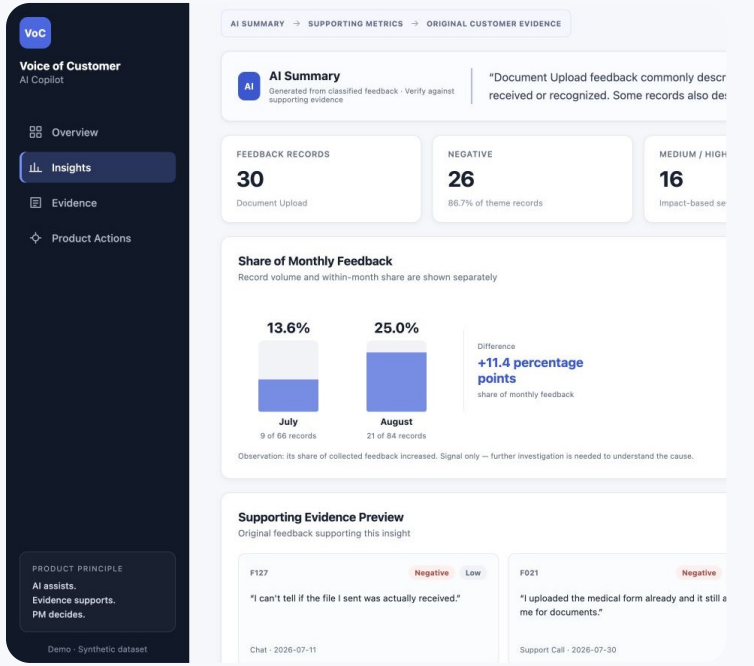
Evaluation changed what I would let the AI decide

Theme	Organize and filter feedback	USE
Sentiment	Add supporting context	SUPPORT
Feedback Type	Review known boundary cases	REVIEW
Severity	Never act as an autonomous priority score	REVIEW

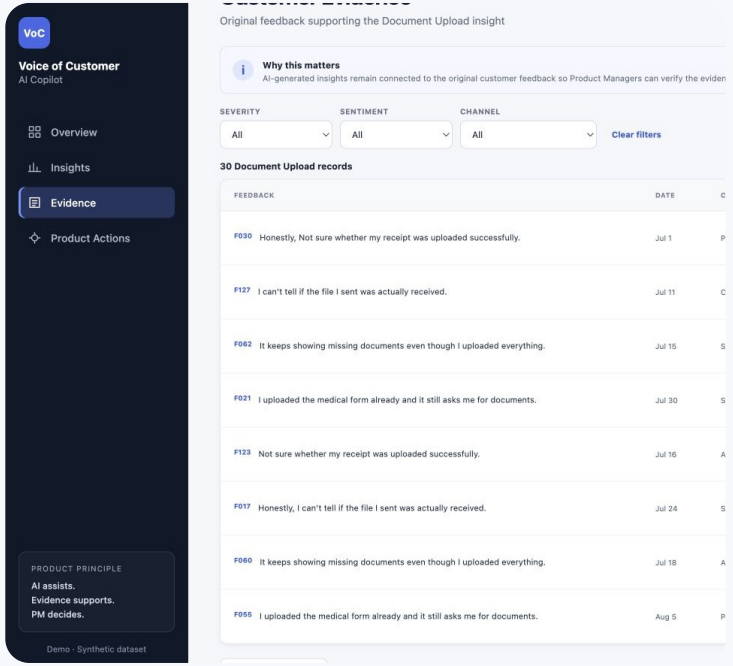
Product recommendation: expose evidence and require PM review when impact affects prioritization.

CASE STUDY

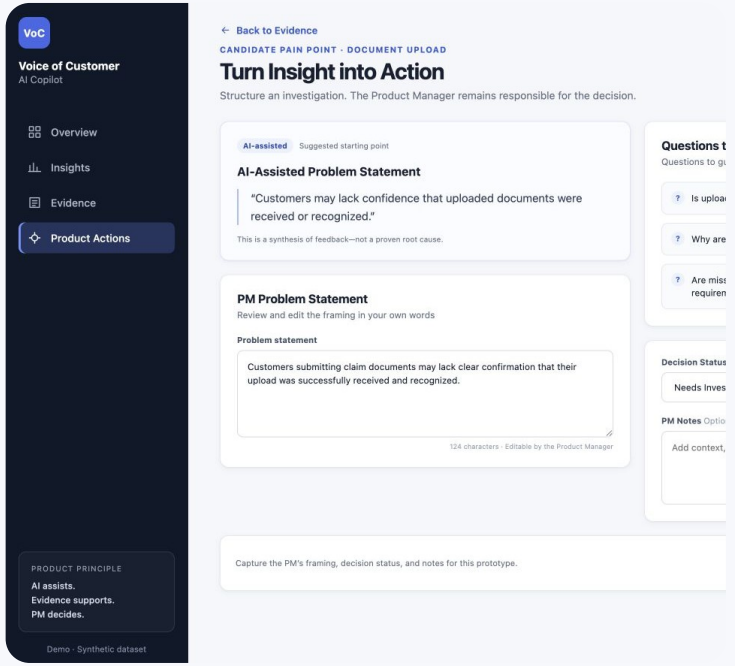
Evidence stays attached to the product decision



CANDIDATE INSIGHT



TRACEABLE EVIDENCE



PM ACTION

The AI proposes a frame. The PM decides whether the evidence supports it.

What I would validate next

CURRENT LIMITS

01 Synthetic feedback, not real customer data

02 24-record coverage-focused holdout

03 Human labels include judgment

04 Prototype taxonomy predates V2

05 No PM usability or outcome study

NEXT VALIDATION

01 Test anonymized real feedback under governance controls

02 Expand and stratify the evaluation set

03 Investigate Severity thresholds and review states

04 Test the workflow with Product Managers

05 Define pilot success metrics before deployment

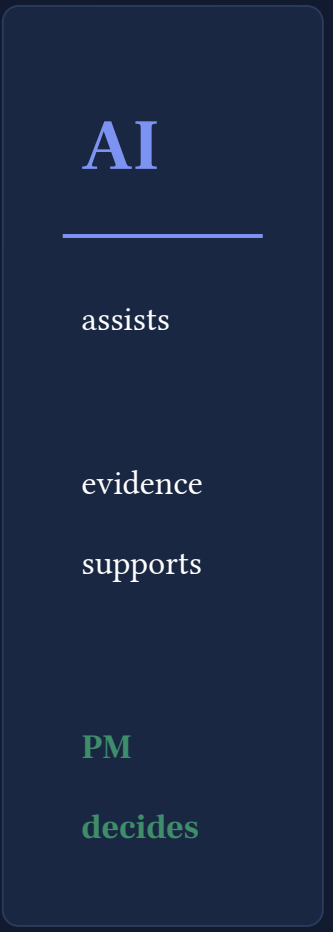
KEY LEARNING

The harder decision was what to trust

01 **Trust Theme for organization**

02 **Treat Severity as a review signal**

03 **Keep evidence and PM judgment in the loop**



Building the classifier produced outputs. Evaluation produced the product decision.

Full Theme taxonomy

Document Requirements

Which documents or paperwork are required

Document Upload

Whether a submitted file was received or recognized

Claim Submission & Form

Entering, editing, saving, or submitting the form

Claim Status

Progress or state of the claim

Reimbursement

Payment timing, amount, or receipt of money

Login & Account

Login, authentication, password, or profile access

Customer Support

Support itself is the primary problem

General Experience

Fallback when no specific Theme applies

Feedback Type, Sentiment, and Severity

FEEDBACK TYPE

- Process Friction
- Technical Issue
- Improvement Request
- Information Need
- Compliment
- General Feedback

SENTIMENT

- Positive
- Neutral
- Negative

SEVERITY

No Issue

No adverse impact described

Low

Limited inconvenience or uncertainty

Medium

Meaningful delay, repeated effort, or intervention

High

Important action blocked or material failure

Severity reflects explicit impact. Emotional wording alone must not increase it.

Dataset and prototype scope

150

synthetic feedback records

July and August 2026

5

raw source fields

ID, date, channel, stage, text

7

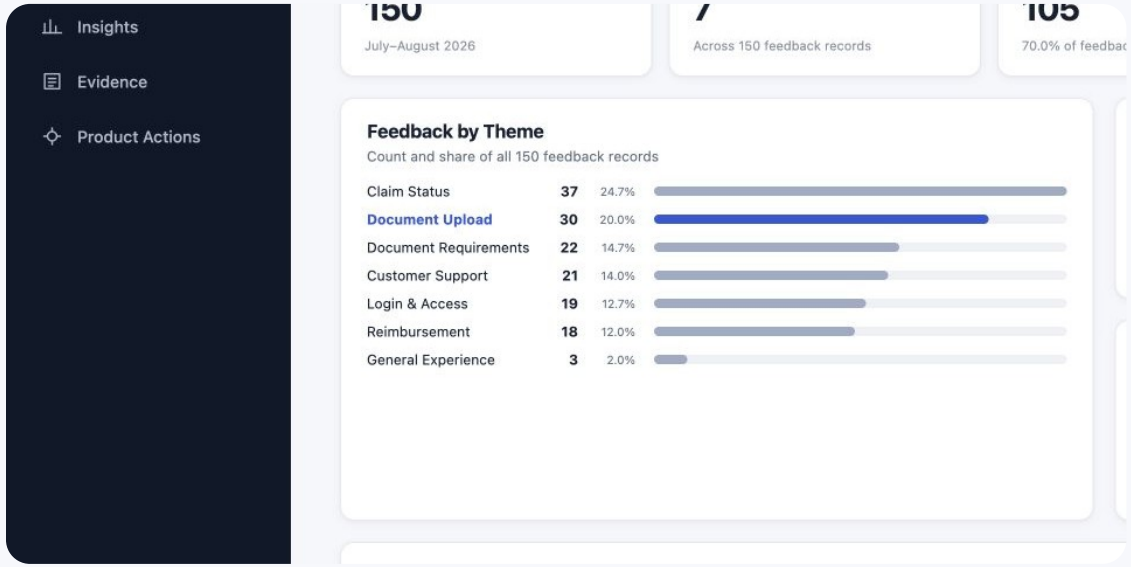
themes in prototype dataset

Earlier taxonomy

0

production integrations

Local interactive demo



Prototype scope caveat

The UI uses the earlier seven-theme labels, including Login & Access and Severity None. It does not display the later Feedback Type dimension. The deck treats it as workflow evidence, not final taxonomy implementation.

Full development comparison

Metric	V1	V2	V3
Theme accuracy	**97.5%**	95.0%	95.0%
Feedback Type accuracy	**85.0%**	90.0%	87.5%
Sentiment accuracy	**82.5%**	52.5%	85.0%
Severity accuracy	**92.5%**	82.5%	85.0%
Exact match	**67.5%**	37.5%	62.5%
Theme macro F1	**86.5%**	84.3%	84.3%
Type macro F1	**80.5%**	85.0%	83.3%

Holdout confusion matrices

FEEDBACK TYPE

	Process	Technical	Request	Info	Compliment	General
Process	3	2	0	0	0	0
Technical	0	5	0	0	0	0
Request	0	0	6	0	0	0
Info	0	0	0	3	0	0
Compliment	0	0	0	0	3	0
General	1	0	0	0	1	0

SEVERITY

	No Issue	Low	Medium	High
No Issue	3	7	0	0
Low	1	3	2	0
Medium	0	0	6	0
High	0	0	0	2

Rows: human reference · Columns: Gemini prediction

Representative holdout errors

HOLD003	**Request for evidence-category examples**	**Severity: No Issue / Low**	Request language triggered impact
HOLD011	**Claim status moved backward after waiting**	**Type: Process Friction / Technical**	System-state wording outweighed process
HOLD015	**Payment arrived; bank description was cryptic**	**Type / Sentiment / Severity**	Mixed factual feedback became praise
HOLD021	**Chat disconnected; user returned to queue end**	**Type: Process Friction / Technical**	Technical event outweighed primary setback

Experiment integrity safeguards

- ✓ Human-reviewed development and holdout references
- ✓ Predictions saved before label comparison
- ✓ SHA-256 locks for frozen artifacts
- ✓ Final candidate selected before holdout
- ✓ Four-field allowlist for model input
- ✓ No post-holdout prompt rerun
- ✓ Structured JSON schema plus local validation

The holdout can inform future work, but it cannot remain an unseen test after that work.

Metrics and technical implementation

Accuracy	Share of records correct for one dimension
Precision	When a label is predicted, how often it is correct
Recall	How many true examples of a label the model finds
F1	Balance of precision and recall
Macro F1	Average F1 across classes with equal class weight
Exact match	All four fields correct for one record

Python ✦ pandas | **Gemini Developer API** | **Structured JSON output** | **Local React prototype**